

L4a: Lattice Trading Rules and Probability of Profit

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Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today we turn a lattice forecast into a precisely defined terminal trading event, compute its probability, and extend the lattice beyond two branches.

Today's concepts

- **Specify a terminal rule:** identify the dated cash flows, holding period, benchmark growth rate, costs, and target.
- **Derive a target probability:** convert a strict return target into a minimum up-move count and evaluate a binomial tail.
- **Generalize the state space:** distinguish terminal and first-passage exits, then represent an N-ary lattice with branch counts.

Lectures and Examples

Lecture: Lattice Trading Rules and Terminal Probabilities

Example: Binomial Lattice Terminal-Target Probabilities

Estimate a binomial lattice from historical prices, convert a strict return target into an up-move cutoff, and evaluate the binomial tail.

Example: N-Ary Recombining Lattice Models

Discretize empirical growth into three branches, construct the recombining states, and plot probability mass at a selected lattice level.

Concept Review: What Trades, Where, and How

1. Security

A **stock** is an ownership claim. An **ETF** is an exchange-traded basket of assets.

Mutual funds are priced periodically, usually once each trading day.

2. Venue

An **exchange** matches buy and sell orders under a common set of market rules.

A market maker helps trading by posting a bid and an ask.

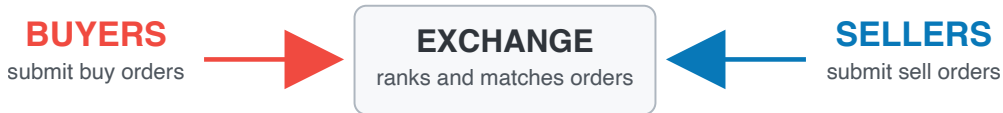
3. Instruction

A **market order** prioritizes execution, but not a specific price.

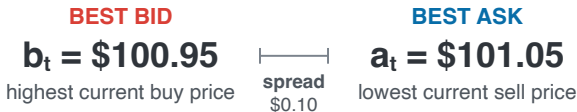
A **limit order** controls the acceptable price, but may not execute.

Execution matters: the order type and current quotes determine the cash flows in a complete trading rule.

The Exchange Matches Buyers and Sellers



TOP OF BOOK



Market sell executes at the bid.

Market buy executes at the ask.

Assume enough size is available at each displayed quote; fees and slippage are omitted.

Market Makers Supply Executable Prices

Citadel Securities is a **market maker**: it quotes prices at which it is willing to buy and sell, giving incoming orders an available counterparty.

One-share example using the preceding quotes

Market buy

$$\text{cash flow} = -a_t = -\$101.05$$

The trader receives one share and pays the ask.

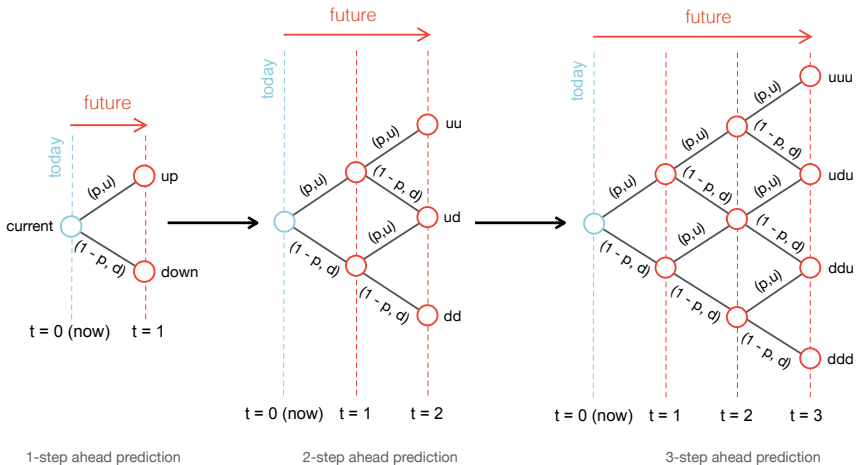
Market sell

$$\text{cash flow} = +b_t = +\$100.95$$

The trader delivers one share and receives the bid.

Modeling consequence: a complete trading rule uses executable entry and exit prices, plus any fees and slippage.

A Recombining Price Lattice



The Binomial Lattice Model

Let $S_0 > 0$ be today's share price. At each step, multiply the price by a constant up factor $u > 0$ with probability p , or a constant down factor $d > 0$ with probability $1 - p$, where $u > d$.

After t steps, let K_t count the up moves. Then the node indexed by $K_t = k$ has price and real-world probability:

$$\underbrace{S_{t,k} = S_0 u^k d^{t-k}}_{\text{node price}}, \quad \underbrace{\mathbb{P}(K_t = k) = \binom{t}{k} p^k (1-p)^{t-k}}_{\text{node probability}}.$$

The model is transparent, but fixed u , d , and p do not reproduce heavy tails or volatility clustering.

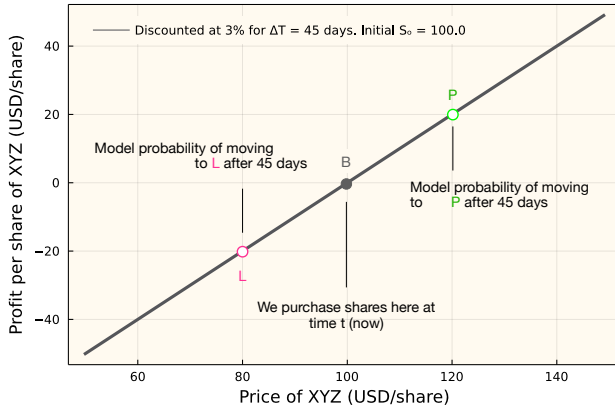
A Forecast Is Not Yet a Trading Rule

A price model describes possible states and their probabilities. A trading rule must add the decisions that define the event of interest.

- When do we enter, and how many shares do we buy?
- Is the exit scheduled at a terminal date or monitored before it?
- Which entry and exit prices represent executable prices?
- What benchmark return and costs must the trade clear?
- Is success defined by a strict or non-strict target?

Without these choices, “probability of profit” is not well defined.

A Scheduled Entry and Exit



Net Present Value of a Long Position

Buy $n_0 > 0$ shares at price S_0 at time 0, then sell all shares at price S_T at time $T = N\Delta t$. Following L1b, let g_y denote the continuously compounded annual growth rate associated with the selected benchmark yield y . The frictionless time-0 NPV is given by:

$$\underbrace{\text{NPV}(g_y, T)}_{\text{time-0 dollars}} = \underbrace{-n_0 S_0}_{\text{entry now}} + \underbrace{n_0 S_T e^{-g_y T}}_{\text{discounted exit}}.$$

Divide by the initial share cost to obtain a dimensionless present-value return:

$$\underbrace{\rho_T}_{\text{scaled NPV}} := \frac{\text{NPV}(g_y, T)}{n_0 S_0} = e^{-g_y T} \frac{S_T}{S_0} - 1.$$

A positive price change alone need not clear the benchmark, spread, and costs.

Short Holding Period

When $|g_y|T$ is small, $e^{-g_y T} \approx 1$. The scaled NPV is then approximately the fractional change in share price:

$$\underbrace{\rho_T}_{\text{scaled NPV}} \approx \underbrace{\frac{S_T}{S_0} - 1}_{\text{price return}}.$$

Under the binomial lattice, $K_N \sim \text{Binomial}(N, p)$ and:

$$\frac{S_T}{S_0} = u^{K_N} d^{N-K_N}.$$

Therefore the $N + 1$ terminal nodes give the complete discrete distribution of the approximate return.

Longer Holding Period

For a longer horizon, or whenever $|g_y|T$ is not small, retain the benchmark discount factor. At terminal node $K_N = k$:

$$\underbrace{\rho_T(k)}_{\text{present-value return}} = \underbrace{e^{-g_y N \Delta t}}_{\text{discount}} \underbrace{u^k d^{N-k}}_{\text{price ratio}} - 1, \quad k = 0, \dots, N.$$

The node probability remains:

$$\mathbb{P}(K_N = k) = \binom{N}{k} p^k (1-p)^{N-k}.$$

The benchmark changes the return attached to each node. It does not change the real-world node probabilities specified by p . When the selected benchmark is risk-free, its one-step accumulation factor uses the L3b notation $R_f = e^{g_y \Delta t}$.

Cumulative Probability of a Terminal Target

Let $\rho_\star > -1$ be a dimensionless target. We ask for the probability that the scheduled terminal return **strictly exceeds** this target:

$$\mathbb{P}(\rho_T(K_N) > \rho_\star).$$

Since $u > d > 0$, the return $\rho_T(k)$ increases with the up-move count k . Therefore, the target event is equivalent to:

$$K_N > \tau(\rho_\star)$$

for a real-valued threshold $\tau(\rho_\star)$. The probability is a binomial tail, not the probability of one terminal node.

Solve the Return Inequality for the Up Count

Begin with the strict target at a node containing k up moves:

$$e^{-g_y N \Delta t} u^k d^{N-k} - 1 > \rho_\star.$$

Move positive terms, take logarithms, and use $\log(u/d) > 0$:

$$k \log(u/d) > \log(1 + \rho_\star) + g_y N \Delta t - N \log d.$$

Thus the real-valued threshold is:

$$\underbrace{\tau(\rho_\star)}_{\text{count threshold}} = \frac{\log(1 + \rho_\star) + g_y N \Delta t - N \log d}{\log(u/d)}.$$

Strict Targets Need the Next Integer

The event is $K_N > \tau(\rho_\star)$. Because K_N is integer valued, the raw strict cutoff is given by:

$$\underbrace{\kappa}_{\text{raw strict cutoff}} = \lfloor \tau(\rho_\star) \rfloor + 1.$$

Encode cutoffs below and above the feasible lattice range by:

$$\underbrace{k_{\min}}_{\text{stored cutoff}} = \min\{\max\{\kappa, 0\}, N + 1\}.$$

Then $\mathbb{P}(\rho_T > \rho_\star) = \mathbb{P}(K_N \geq k_{\min})$. A non-strict event uses a different raw cutoff when a node lies exactly on the target.

Threshold Edge Cases

The stored cutoff $k_{\min} \in \{0, \dots, N + 1\}$ makes all edge cases explicit.

- If $k_{\min} = 0$, every feasible terminal node succeeds, so the probability is 1.
- If $1 \leq k_{\min} \leq N$, sum the binomial probabilities from $k_{\min}$ through N .
- If $k_{\min} = N + 1$, no feasible terminal node succeeds, so the probability is 0.
- At fixed N , increasing g_y raises the threshold and cannot increase the tail probability.
- Changing the horizon changes both the threshold and the distribution of K_N . Recompute the full probability.

Implementation: for a strict tail, use the complementary binomial CDF at $k_{\min} - 1$ only after handling the two edge cases.

Terminal and First-Passage Rules Differ

The binomial tail concerns liquidation at the scheduled date T . A take-profit or stop-loss rule monitored at each step depends on the **first** boundary crossing and is path dependent.

- Two paths can end at the same terminal node but cross a boundary at different earlier times.
- Propagate only probability mass that has not already crossed a boundary, then record the absorbed mass at each first crossing.
- A stop order becomes a market order after its trigger and need not fill at the trigger price. A limit order may not fill.

Terminal node probabilities alone cannot evaluate a monitored exit rule.

N-Ary Lattice Models

Replace the two branch factors with m positive factors $f_1, \dots, f_m$. Let branch j occur with probability $p_j \geq 0$, where $\sum_{j=1}^m p_j = 1$.

After t steps, a recombining node is indexed by a branch-count vector:

$$\mathbf{x} = (x_1, \dots, x_m), \quad x_j \in \mathbb{Z}_{\geq 0}, \quad \sum_{j=1}^m x_j = t.$$

Order does not matter because the branch factors are constant and multiplication commutes. Paths with the same count vector share one node.

N-Ary Node Values

For count vector $\mathbf{x}$ at level t , the node price is:

$$\underbrace{S_t(\mathbf{x})}_{\text{node price}} = S_0 \prod_{j=1}^m f_j^{x_j}.$$

Independent branch draws give the multinomial node probability:

$$\underbrace{\mathbb{P}(\mathbf{X} = \mathbf{x})}_{\text{node probability}} = \frac{t!}{x_1! \cdots x_m!} \prod_{j=1}^m p_j^{x_j}.$$

Distinct count vectors can still produce the same price if the factors have additional multiplicative relationships.

More Branches Create More States

The number of nonnegative count vectors at level t is the stars-and-bars count:

$$\underbrace{L_t}_{\text{nodes at level } t} = \binom{t + m - 1}{m - 1}.$$

The total number of nodes through height h is:

$$\underbrace{\sum_{r=0}^h L_r}_{\text{all stored nodes}} = \binom{h + m}{m}.$$

More branches give a finer discrete one-step distribution, but also increase the state count. They do not automatically improve out-of-sample forecasts.

N-Ary Example: Discretize, Build, Check

The companion notebook uses empirical log growth rates for one firm.

1. Divide the one-step growth sample into three equal-width bins.
2. Estimate one representative factor and one frequency for each bin.
3. Build each recombining state from its three branch counts.
4. Compute its multinomial probability and verify that all node probabilities at the selected level sum to one.
5. Plot price and annualized multistep log growth against probability mass.

Example: [N-Ary Recombining Lattice Models](#). The histogram of node locations is not a probability distribution unless each node is weighted by its probability mass.

Optional Advanced Material

Advanced: [First-Passage Exit Rules](#)

Propagate only probability mass for positions that remain open, then compute take-profit-first, stop-loss-first, and still-open probabilities.

Advanced: [Execution-Aware Probability of Profit](#)

Add the bid–ask spread, fees, and slippage to the terminal rule, then compare its probability with the frictionless result.

Optional; neither notebook is a prerequisite for L4b.

Summary

Today we converted a lattice forecast into a complete terminal target event, derived its exact probability, and generalized the state space.

Key takeaways

- **Profit is a cash-flow event**: entry, exit, horizon, benchmark, costs, and target strictness must be specified.
- **Monotone terminal targets become count tails**: solve for the integer boundary and handle thresholds outside the feasible range.
- **The state depends on the rule**: monitored exits need path information, while N-ary terminal states use multinomial branch counts.

Next, we take the continuous-time limit and introduce stochastic price models.

Today: use the lattice to define the event before computing its probability.